

# Spreading 32 Points in the Plane

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## Abstract

We place 32 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 29.22375+. Erich Friedman’s table of best known configurations lists  $n$  up to 31. Deleting a point from our 33-point configuration gives 29.80601; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

## 1 Result

**Theorem.** *There are 32 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 29.22375\dots$$

The exact value, a rational number, is given in Appendix A.

## 2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for  $n \leq 31$ . The value for  $n = 32$  is not on the page. Deleting a point from our 33-point configuration gives 29.80601; this one is smaller.

## 3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n32.txt
```

in <https://github.com/Ge-limin/math719>.

## 4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

## A Exact value

```
numerator:
29223754865599733656081498
denominator:
999999999998213138302733
```

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\*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

## B Coordinates

32 points,  $x$  and  $y$  per line. File: friedman-maxmin2d/coords/n32.txt.

```
-3.356408058484 -0.682444621017 -0.818275262118 1.483542034665
-2.701326562891 -1.438002847848 -4.253718609167 1.500631567416
-1.279990871134 0.596514021358 -2.562810272840 -2.471822029009
-3.477745116341 -2.068220453012 -1.702585921698 -1.488173871641
-4.534038517572 -0.396676960593 -0.342356634737 -0.290695394864
-0.595605582013 -2.224443288715 0.580217857492 -0.676514402026
-2.642263315666 2.005103658061 -0.022858954538 2.089605478238
-2.750558901876 0.398078928947 -1.336153314299 -0.401907623028
0.724224658271 0.313062296027 -3.287889890761 1.241450382332
0.597066473622 1.304944746762 -3.608092034073 2.264284843145
-0.847800410186 2.654823661557 -1.643216717766 2.048760217983
-0.198349833959 0.698881303189 -2.180547706651 2.892131671368
-4.674400404548 0.593423307403 -0.742942359717 -1.206954679643
-3.738268000690 0.241775582472 0.179632132513 -1.592773686805
-1.562831646174 -2.478360087700 -4.132826611935 -1.312662226181
-2.104932326782 1.161732204676 -2.330364088312 -0.509354990770
```